

PORTFOLIO

Wenhou Yan

2024 - 2026



Wenhou Yan (Tomic)

Product Designer (Industrial/UI/UX design)

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EXPERIENCE

Sunswift Solar Racing Team

Product Design & Manufacturing Support

2025.04 – 2025.07

Independent Additive Manufacturing Studio

Founder

2025.09 – Present

EDUCATION

2025/2 – 2026/5

University of Sydney (USYD)

Master – Interaction Design and Electronic Arts

2021/9 – 2024/7

University of New South Wales (UNSW)

Bachelor – Industrial Design

SKILLS

CAD Modeling

Rendering

3D Printing

Presentation

Communication

Storytelling

User Research

Market Research



PROJECTS



#01
Sangre
Home Health Devices & Smart Living Agents



#02
Bambino V2
Brief description here



#01

Sangre

SANGRE is a home cardiovascular monitoring device designed to simplify blood testing, reduce user anxiety, and support long-term health management through an approachable and user-friendly experience.

DURATION

14 weeks

ROLE

Lead Designer

TOOLS

Solidworks, KeyShot, AI,figma

RESEARCH

Convenience

The user-friendly interface and simple operation process make it easy for users to perform lipid tests at home and get results quickly.

Compactness

Design small, portable devices that are easy to store and carry, suitable for use at home.

Ease of Maintenance

Design equipment that is easy to clean, calibrate and maintain to ensure long-term stability and accuracy while reducing the operational burden on users.

Ease of Use

Ambient Presence

Scalable Information Density

Fine-matte plastic

Foldable Screen

Target group

Chronic Illness in Young Adults

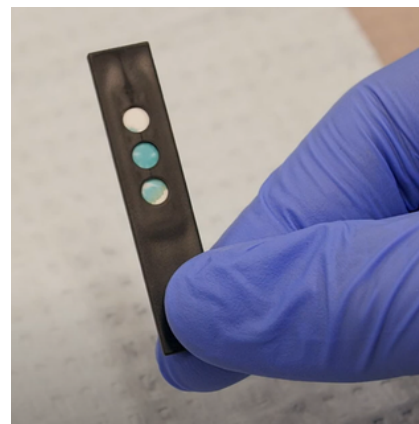
A home-use, easy-to-operate, aesthetically pleasing multifunctional blood testing device focused on the measurement of the four main blood lipid parameters (Total Cholesterol, High-Density Lipoprotein Cholesterol, Low-Density Lipoprotein Cholesterol, Triglycerides), with additional capabilities for uric acid and glucose testing.

PRIMARY RESEARCH CONGENERIC PRODUCT



CARDIO CHECK

A field detection tool that is widely used in a variety of clinical Settings, especially in the management of cardiovascular diseases and related diseases. Its fast, efficient and portable characteristics make it widely recognized and used in the medical field.



ACCUTREND PLUS

This portable device is capable of measuring a variety of parameters, including total cholesterol, triglycerides and blood sugar, for diabetes and cardiovascular disease management.

CHOLESTECH LDX™ANALYZER

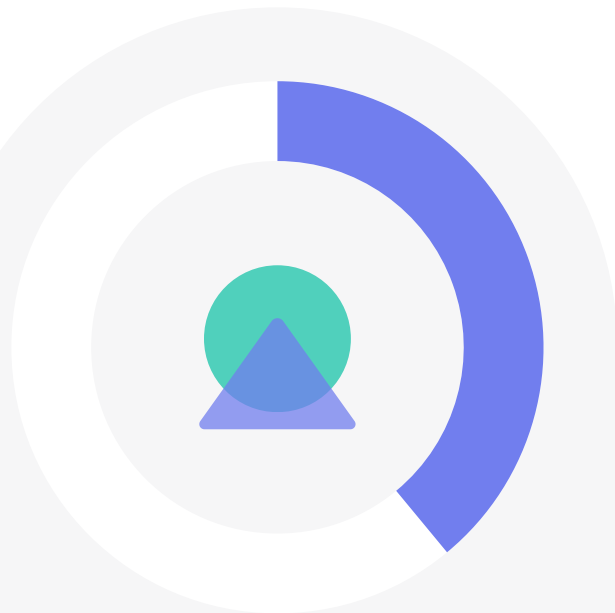
This device is used for the rapid detection of blood lipids and glucose, and is widely used in clinical and field screening. Its fast, accurate results make it a serious competitor to the CardioChek PA.



DESIGN CONCEPT

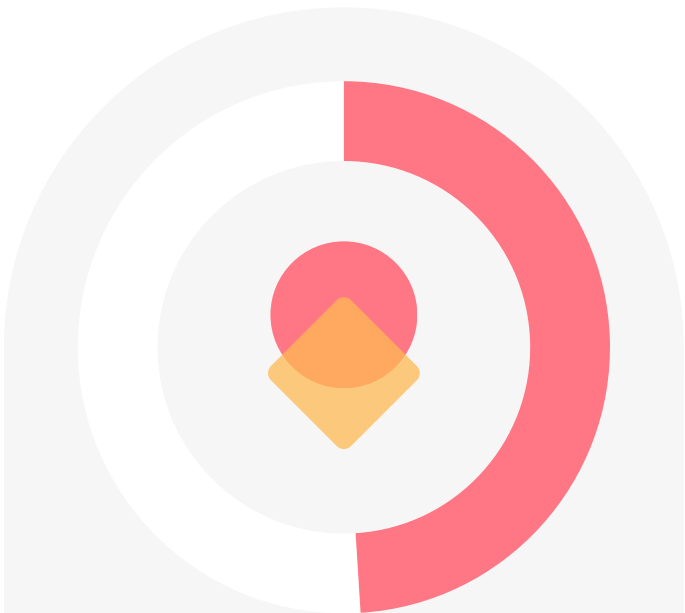
DESIGN BRIEF

A home-use, easy-to-operate, aesthetically pleasing multifunctional blood testing device focused on the measurement of the four main blood lipid parameters (Total Cholesterol, High-Density Lipoprotein Cholesterol, Low-Density Lipoprotein Cholesterol, Triglycerides), with additional capabilities for uric acid and glucose testing.



Reliability

Ensuring that equipment can provide accurate measurement results comparable to professional laboratories is essential for user trust and effective health management.



Convenience

The user-friendly interface and simple operation process make it easy for users to perform lipid tests at home and get results quickly.



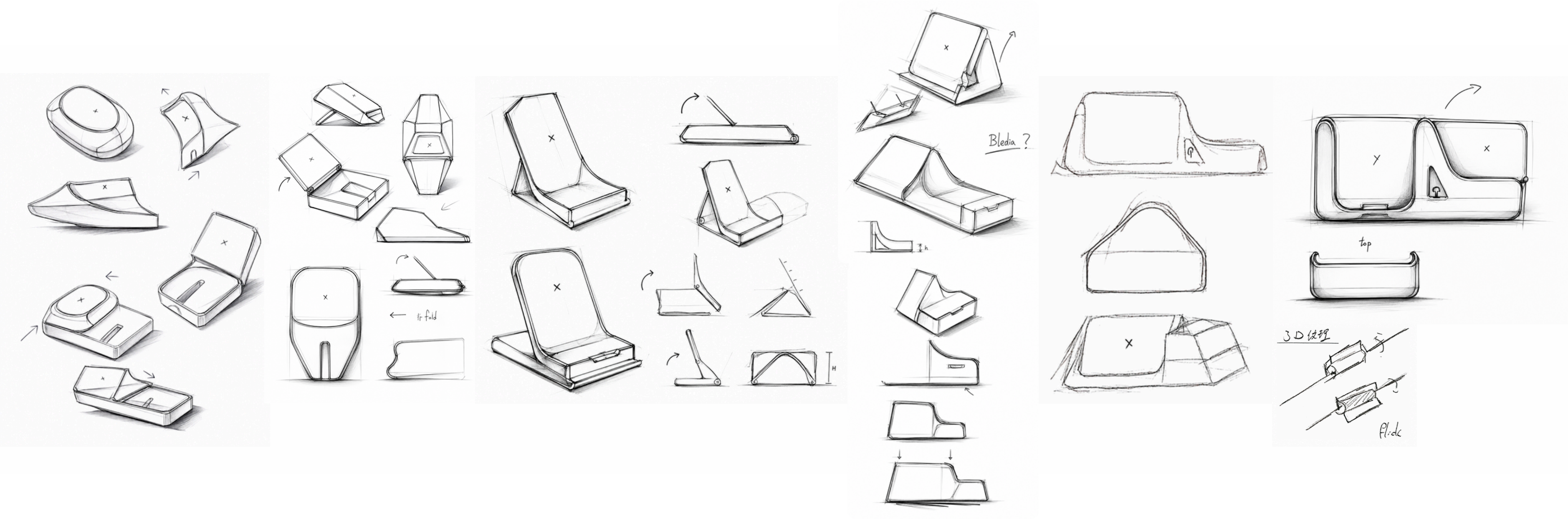
Compactness

Design small, portable devices that are easy to store and carry, suitable for use at home.



Ease of Maintenance

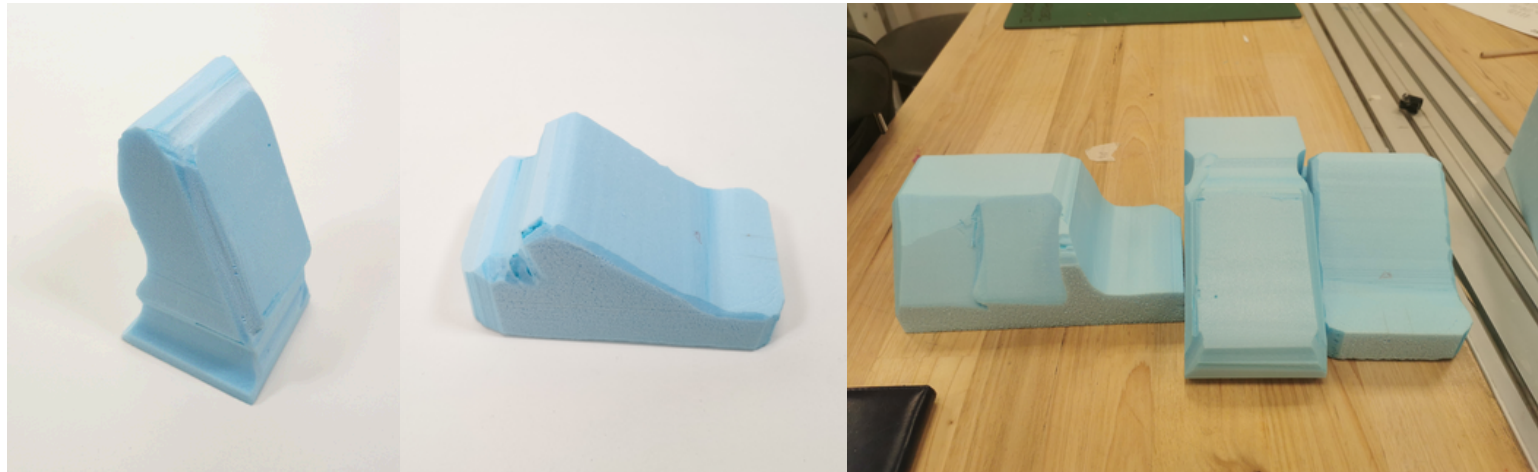
Design equipment that is easy to clean, calibrate and maintain to ensure long-term stability and accuracy while reducing the operational burden on users.



Mechanism & Form exploration.

Early ideation sketches investigated user ergonomics, usability, and visual impact across varying forms. These initial studies specifically explored alternative configurations for tool and accessory storage, as well as adjustments to screen viewing angles and multi-directional orientations.

Prototyping, Iteration & Evaluation.



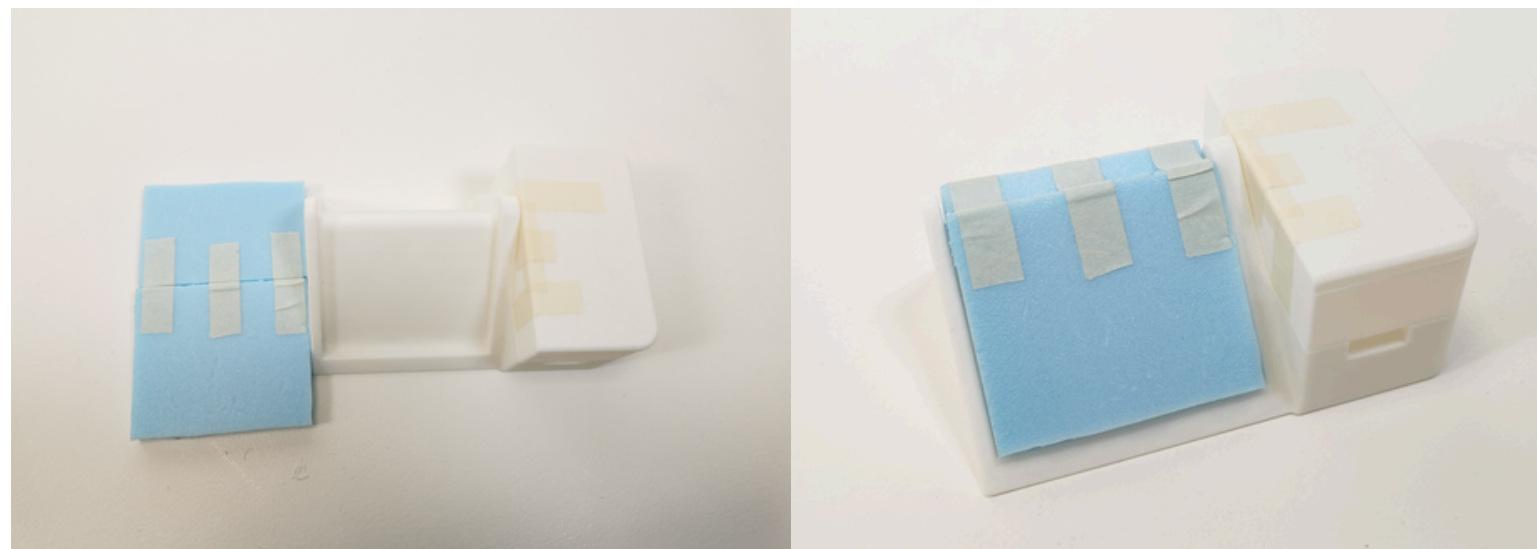
#01

An early-stage volumetric model used to define the rough aesthetic and silhouette direction. While the surfaces and fine details remain coarse and unrefined, the overall form-factor closely approximates real-world proportions.



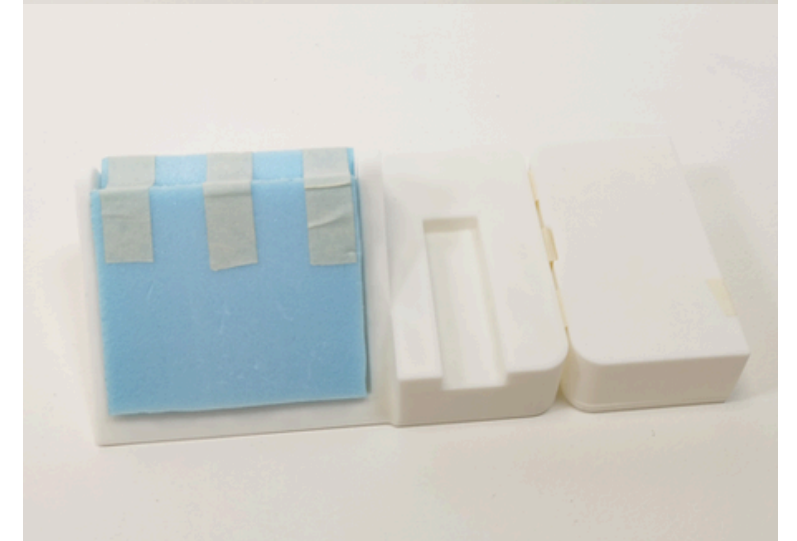
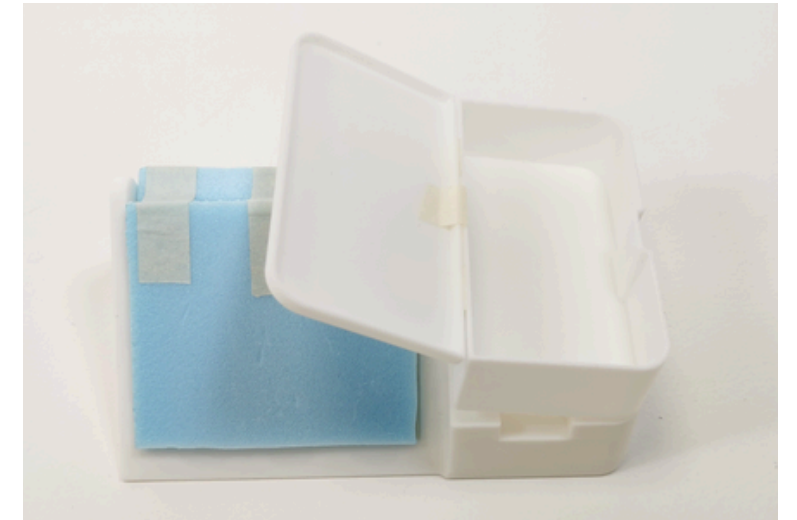
#02

A refined second-round prototype developed by filtering and optimizing the insights gained from the Batch 01 rough models. This iteration features a significantly higher level of fidelity, surface finish, and detail precision.



#03

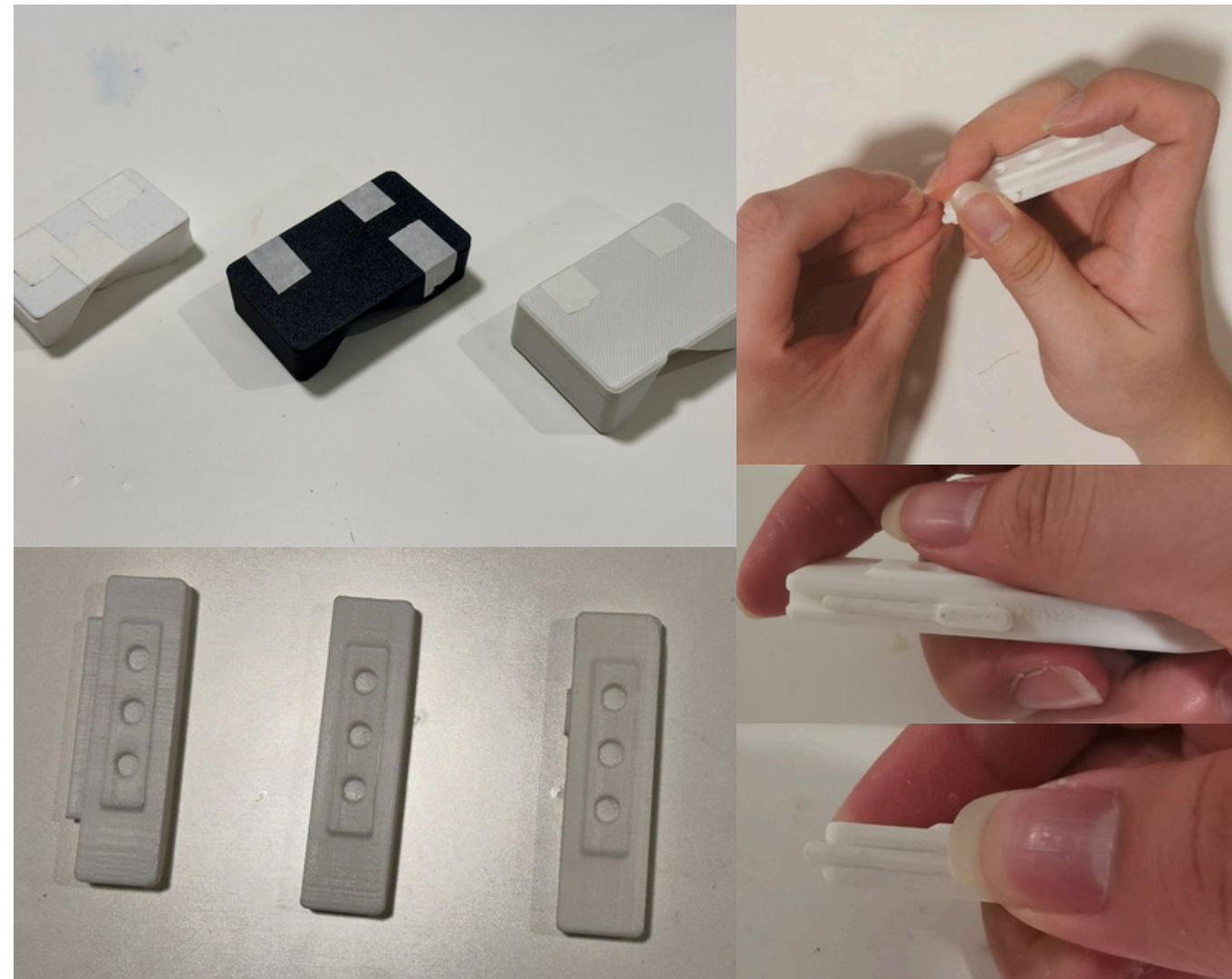
In Batch 02, the model on the right features a relatively lower profile, making it significantly easier to package and transport. Consequently, during the third phase, this right-hand concept was selected for further refinement and 3D-printed to validate its visual aesthetics and form factor.



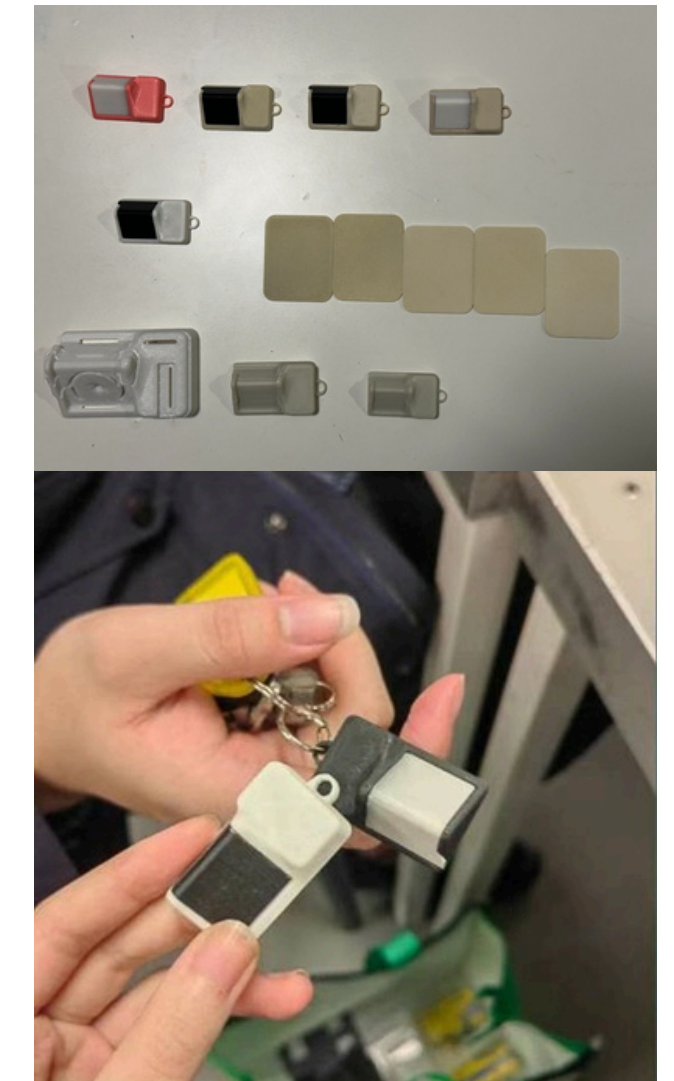
Supplementary components and auxiliary parts utilized in alternative testing phases.



An engineering flaw identified during small-scale vacuum forming (blister packaging) tests, where the draft angle indicated by the arrow was far too steep, potentially causing part deformation or binding during ejection. This issue was successfully corrected in the final production model.



Design testing for the lateral specimen storage compartment alongside various test strip holder configurations. The objective is to seamlessly integrate the lancing device and test strip housing into a single, cohesive structure, thereby eliminating the risk of components becoming misplaced, scattered, or lost.



Originally created as small-scale models purely for paint and color testing, these pieces garnered unexpected interest from a wide range of people. Capitalizing on this, they were slightly modified and repurposed into functional keychain charms.

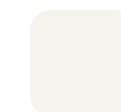


HELLO, SNGRE

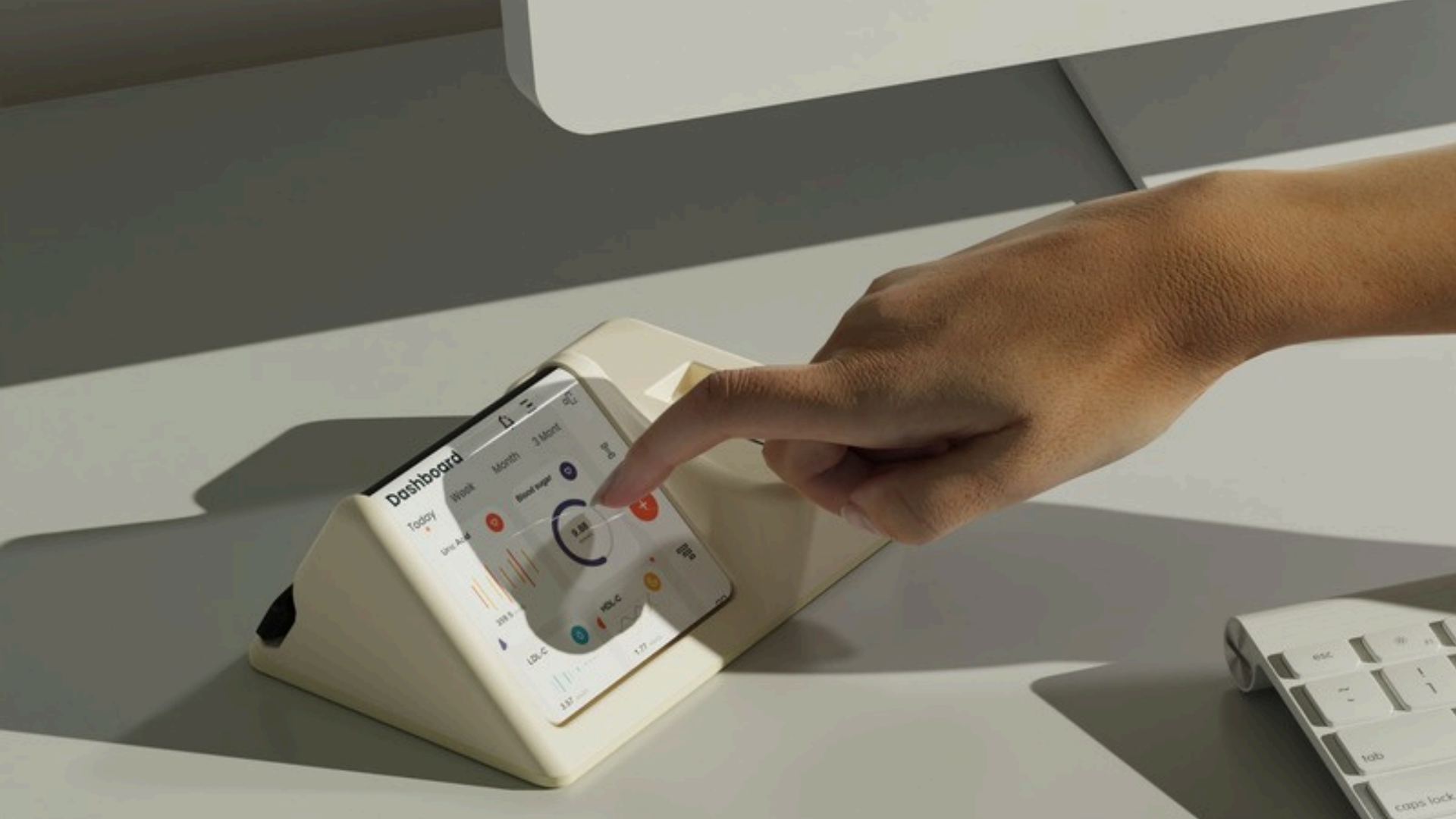
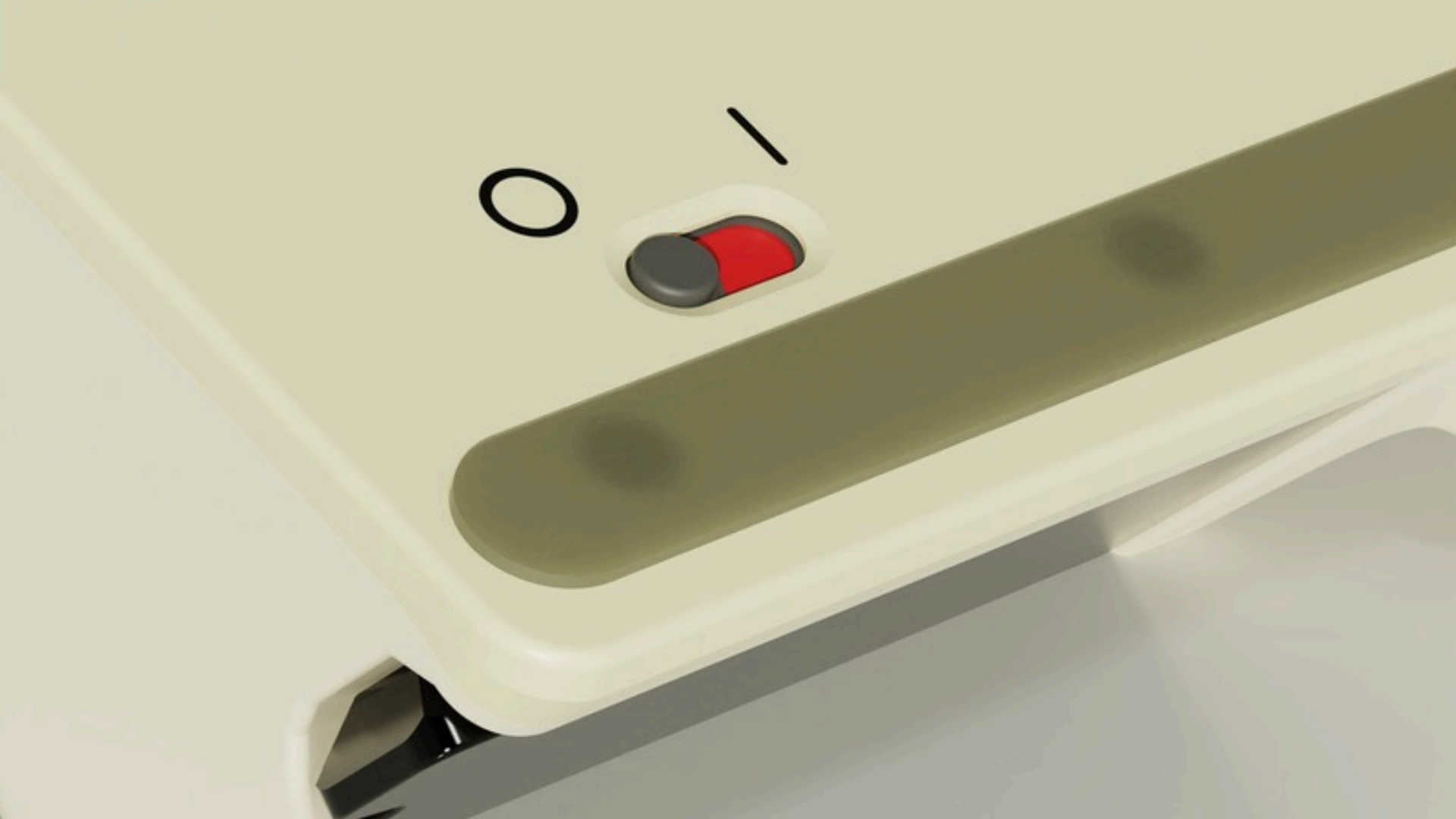
A 1:1 scale model featuring all functional operational movements except for the actual electronic testing capabilities, with components meticulously split according to real-world manufacturing and assembly logic. The model includes three distinct screen variants to represent different states: a folded configuration, an active usage mode, and a version equipped with a fully articulated mechanical hinge.



- #deddd0



- #f4f3ec



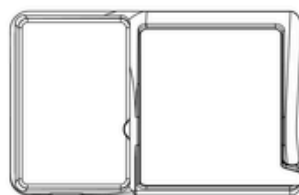
Design for Manufacturing.

The core architecture is primarily designed around injection molding, while integrating cold-pressed aluminum for select internal structures to simplify production and lower manufacturing costs.

Leveraging raw materials that are highly prevalent in the consumer electronics market. ABS injection molding offers low production costs, manufacturing efficiency, and mature surface treatment methods. Compared to typical metal materials, it is significantly easier to sanitize and provides a much more skin-friendly tactile experience.

- Utilizing a top-and-bottom split shell design, the assembly strategically uses its outer edges to conceal component seams while incorporating a distinctive two-tone color blocking scheme.
- A bottom-heavy component layout concentrates the primary electronics and heavier elements in the lower section of the device, allowing the upper housing to be easily detached.
- Powered by a standard, fully removable 18650 lithium battery to ensure longevity and ease of service.
- The screen assembly operates via wireless power delivery. As an added functional benefit, this system can also be utilized to supply wireless power to a smartphone if needed.

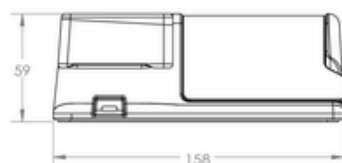




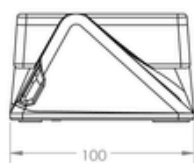
Top view



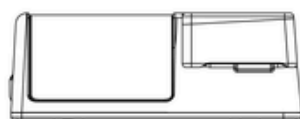
Right side view



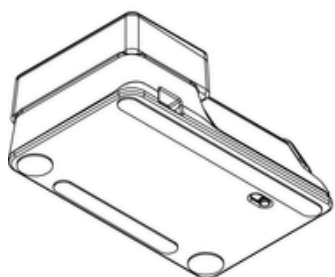
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Back view



Left side view



Front view



Bottom view

NO.	part name	material	amount
1	body part	PP	2
2	18650	battery	1
3	screen	aluminum	1
4	sensor	304	1
5	switch	PP	1
6	Test strip storage box	PET	1

NAME: Wenhou Yan

Industrial Design

Description:

CardioCheck
Sangre

Capacity

File Name: SANGRE Z5319444

Technical drawing of a mechanical part, showing multiple views and dimensions.

Views:

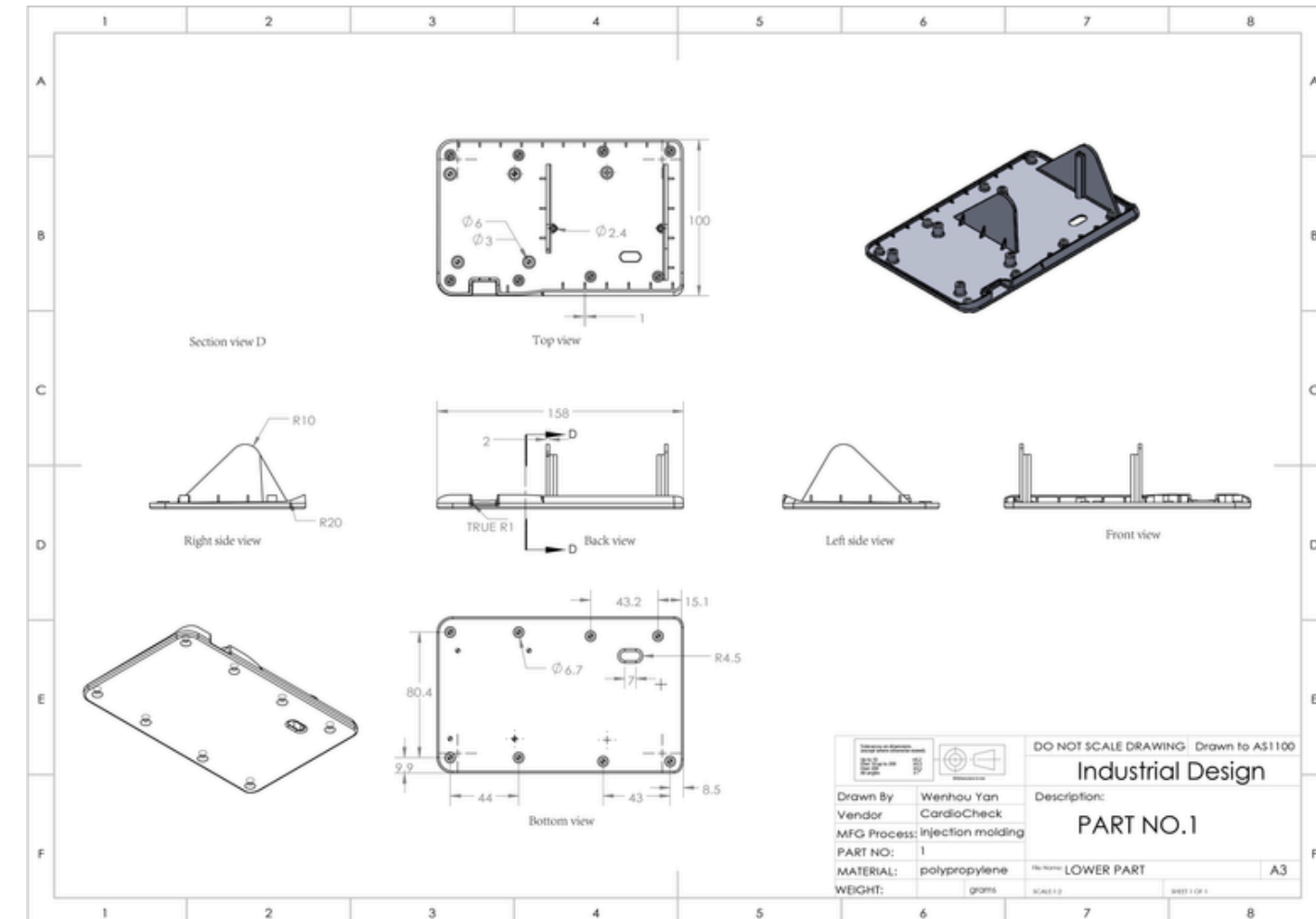
- Front view
- Top view
- Back view
- Left side view
- Right side view
- Bottom view
- SECTION A-A

Dimensions:

- Front view: 7.5, 11.1, 20, R1
- Top view: 135.00°, 3.00"
- Back view: 5.87", 18.4", R1
- Left side view: 283°, 35.29°, 41.3°, 11.3°, R6
- Right side view: 95.7°, A
- SECTION A-A: 78, 10, 5.9, 1.500, 24.500, $\phi 5$

Title Block:

Drawn By: Wenhou Yan Vendor: CardioCheck MFG Process: injection moulding PART NO: 2 MATERIAL: polypropylene WEIGHT: grams		DO NOT SCALE DRAWING Drawn to AS1100 Industrial Design PART NO.2 TITLE: UPPER PART SCALE: 1:2 SHEET 1 OF 1
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#02

Banbino V2

Based on the market position of the Breville Bambino / Bambino Plus, the goal is to address core user pain points.

DURATION

12 weeks

ROLE

Individual Event

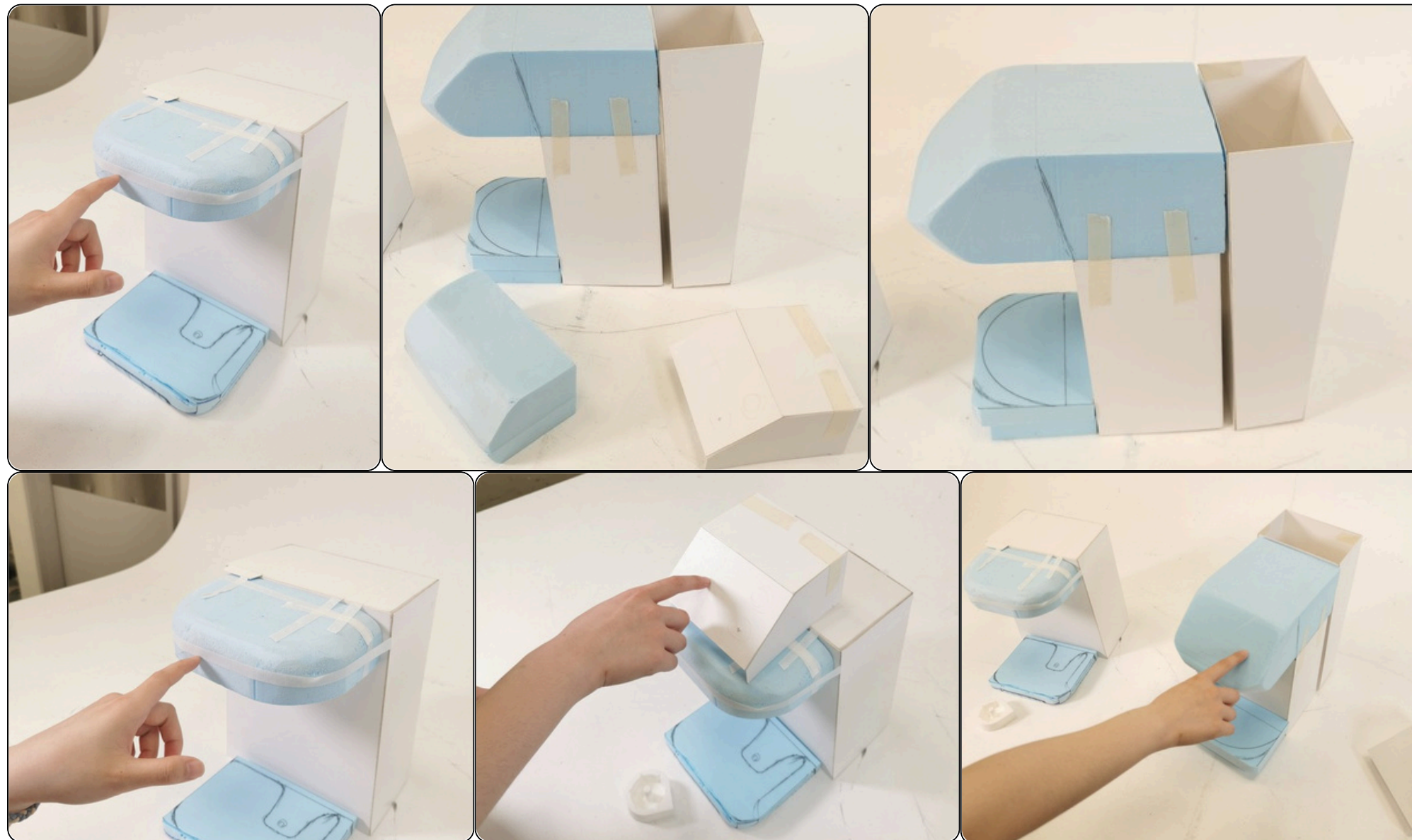
TOOLS

Solidworks, Rhino, KeyShot,

PRODUCT POSITIONING & TARGET GROUPS

- The "Ultra" iteration broadens the Bambino's target demographic by introducing scalable interaction complexity. For users seeking efficiency, the machine retains the intuitive, streamlined experience of the original Bambino. Conversely, for enthusiasts, the integrated adaptive haptic feedback—delivered via a precision-engineered knob—transforms the extraction process into a deliberate sensory ritual, significantly heightening user engagement.
- Furthermore, the Ultra re-engineers the mechanical locking logic of the portafilter. By optimizing the force distribution during the bayonet mount engagement, the redesign minimizes the rotational torque that typically causes machine displacement. This structural refinement effectively resolves the stability issues inherent in traditional suction-cup-based fixation, resulting in a more grounded and premium tactile experience.
- **Enhanced Group Head Mechanism**
- An optimized, easier-to-lock group head architecture that counteracts the machine's tendency to twist or rotate during engagement, achieved seamlessly without introducing unnecessary mechanical complexity to the overall design.
- **Dynamic Visual Interaction**
- Integrating a much more intuitive and vivid visualization feedback system to drive the user interface. This layout fundamentally transforms the interaction experience, replacing the monotone and uninspired feel of the original button-based interface with a dynamic, engaging workflow.



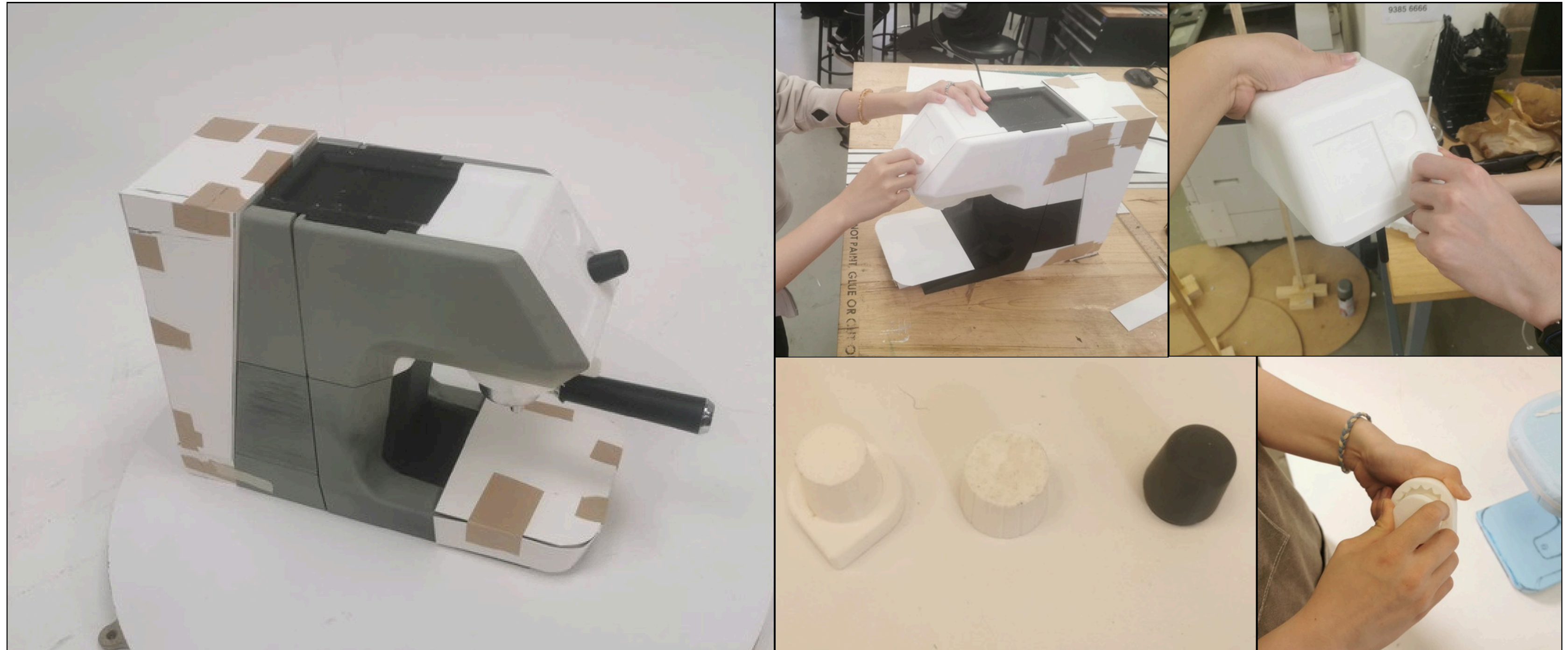


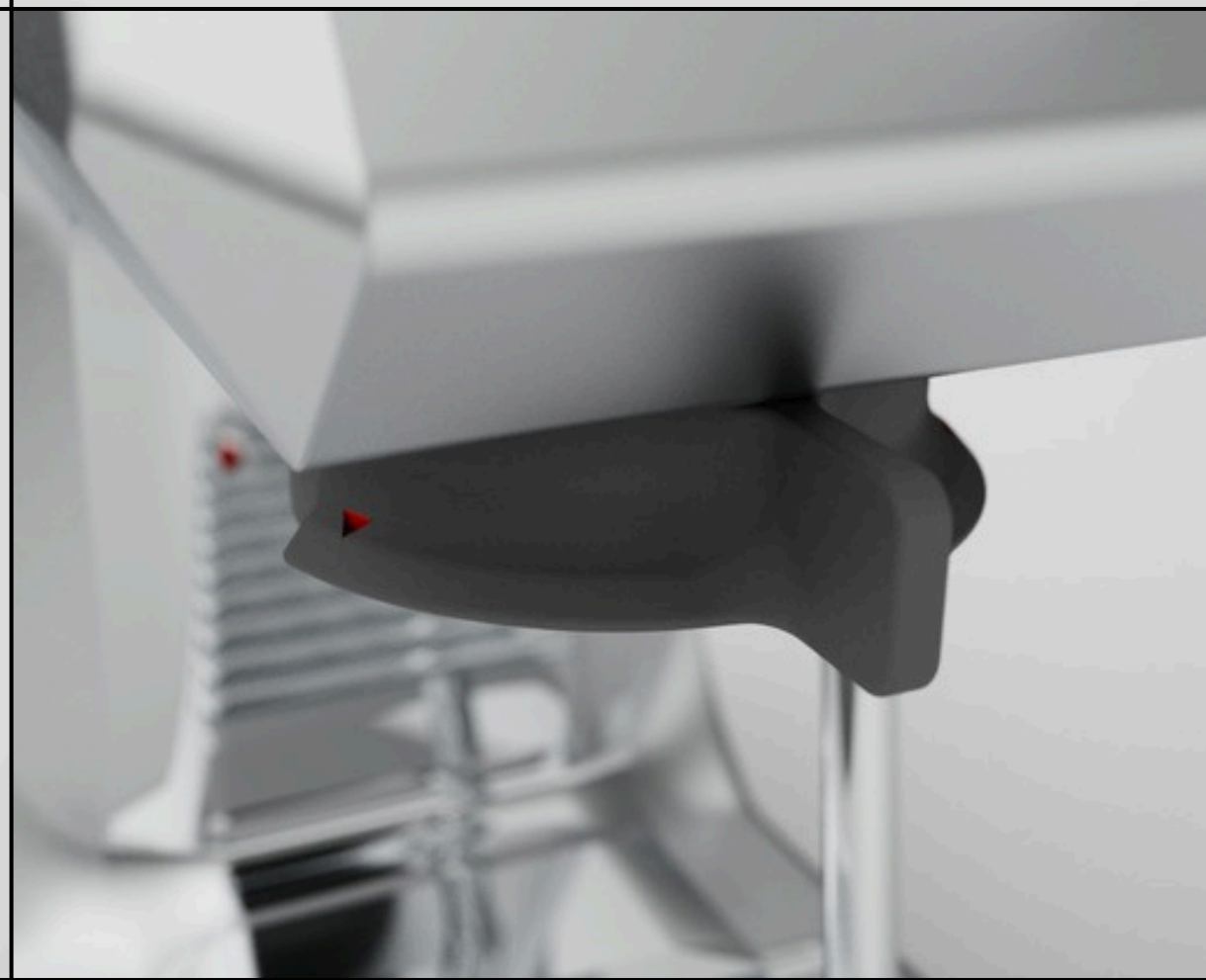
- The rough models used in the early stage for user usability testing were approximately 1:1 in scale, used to confirm the flow of actions and user usage habits. They could also intuitively simulate the overall user experience, such as the overall height of the machine, the angle of the screen, and the position of the screen, etc., for the overall design.

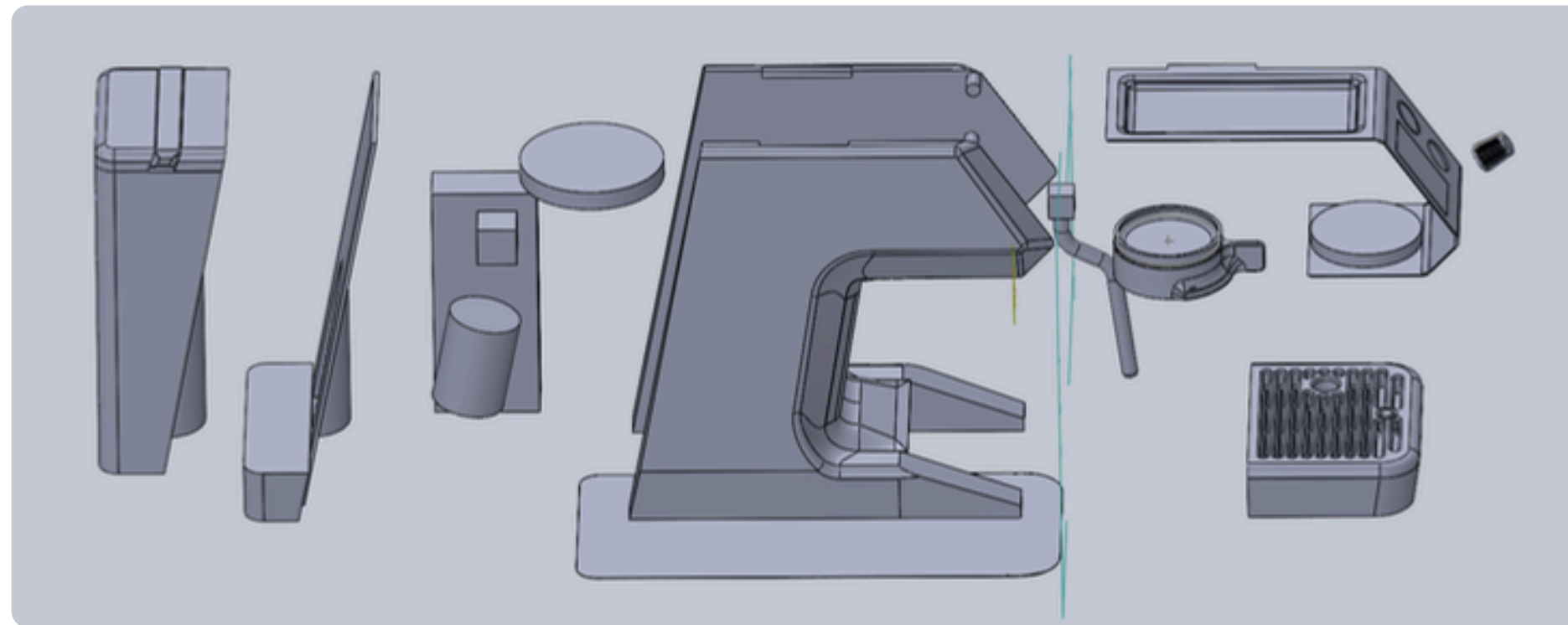
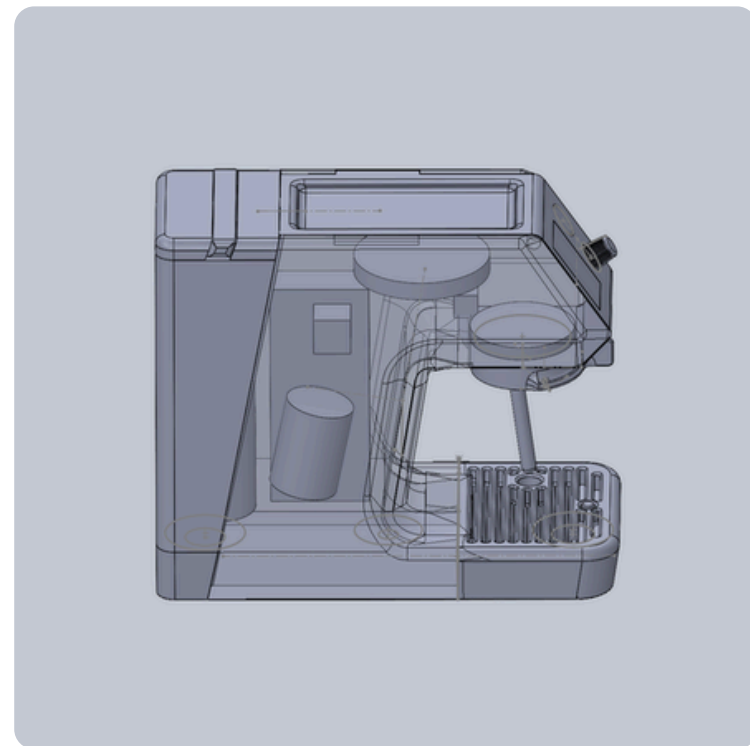
- A detail test model (or isolated prototype component) created by splitting a highly detailed feature from the main body and manufacturing it as a standalone piece. This targeted approach accelerates design iteration cycles and significantly reduces material waste and prototype fabrication costs.



PHYSICAL MODEL







- Unlike the original prototype, which was limited to a single brushed stainless steel finish, this product incorporates a diverse palette of colorways into its exterior design to offer greater aesthetic variety.

Get in touch!

If you are interested in this project or potential collaboration, feel free to reach out.

Contact

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Phone: +61 0449923613

Wechat:

